

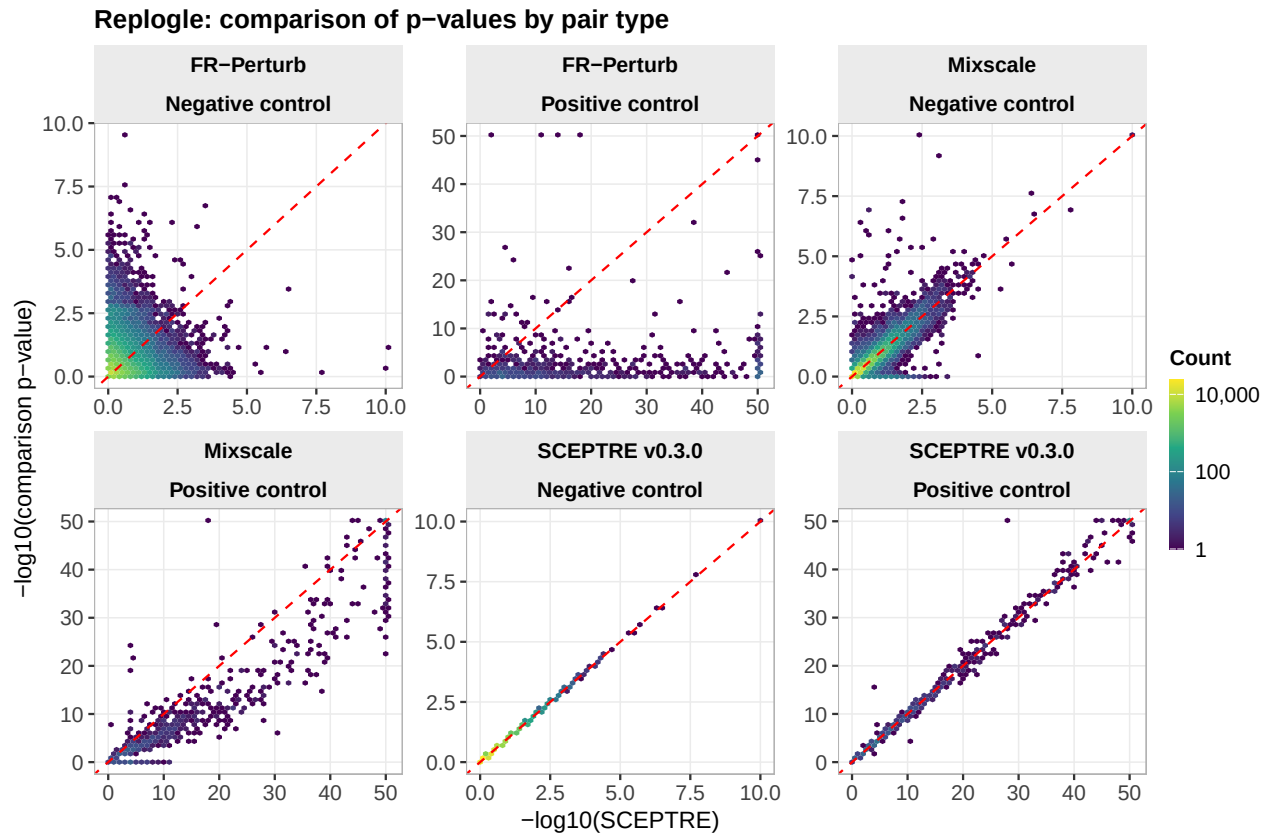
meeting 2026-09-10

2026-09-10

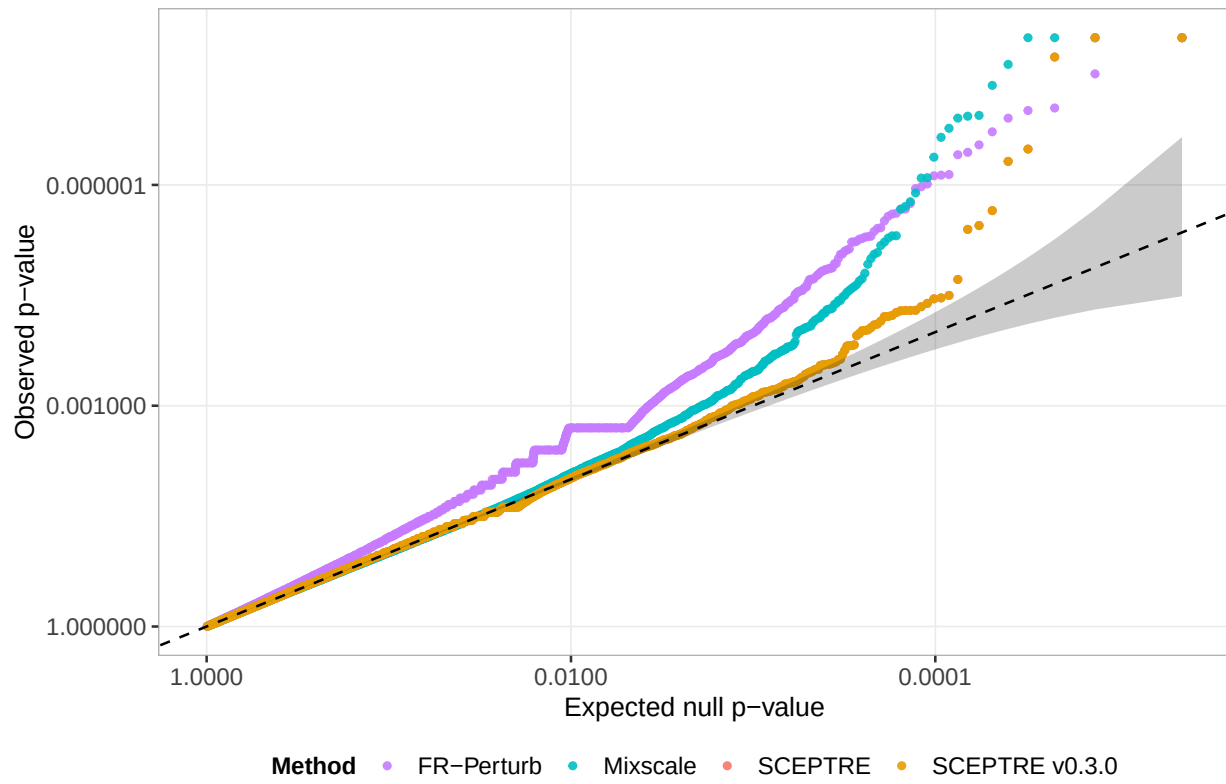
Association

First, we show the positive and negative control analysis with the two legacy SCEPTRE versions added. The only substantial difference is that `sceptre-manuscript` (high MOI only) only allows p-values as small as $2e-16$, hence the pairwise scatterplot with current SCEPTRE not looking good, but the rank-based assessments are fine.

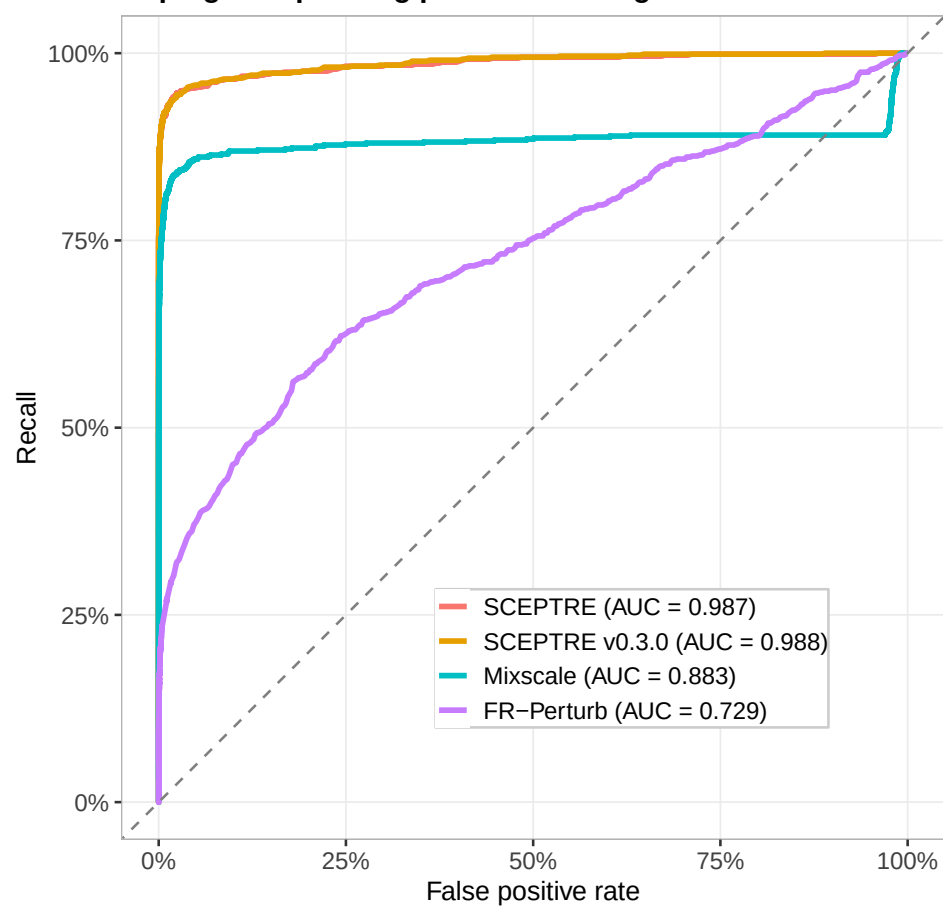
Replegle: comparison with SCEPTRE v0.3.0 from 2023



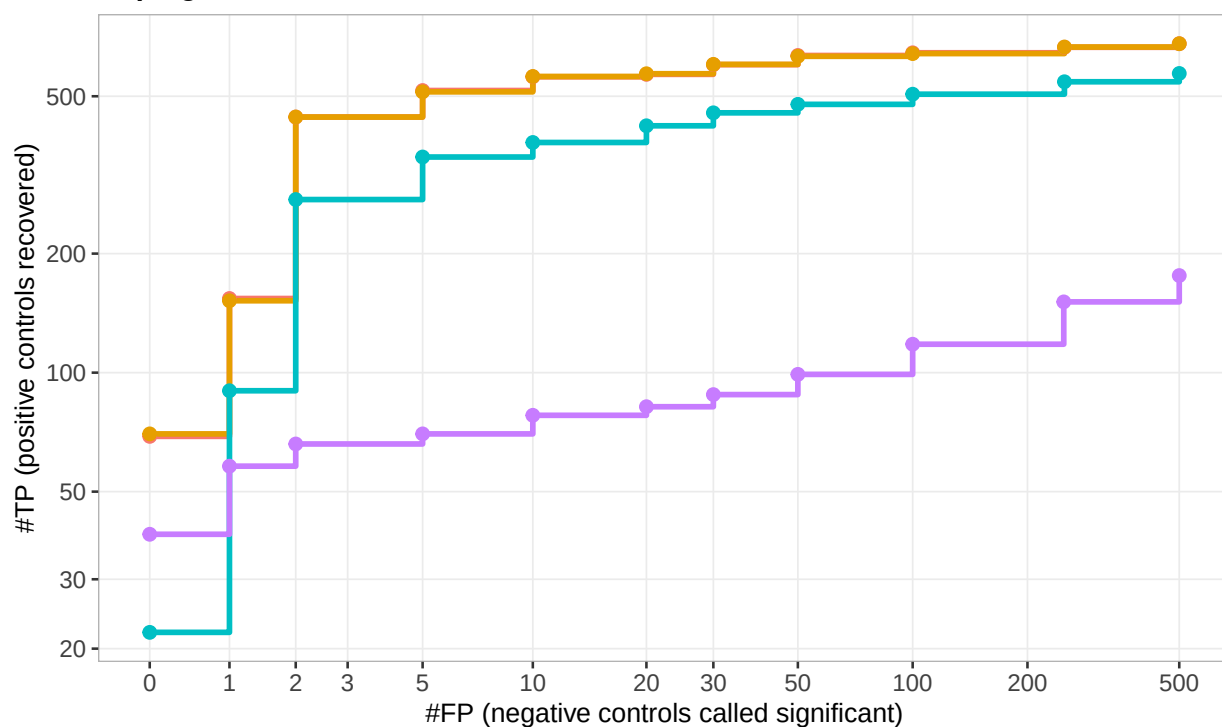
Replogle: miscalibration in negative controls



Replogle: separating positive and negative controls



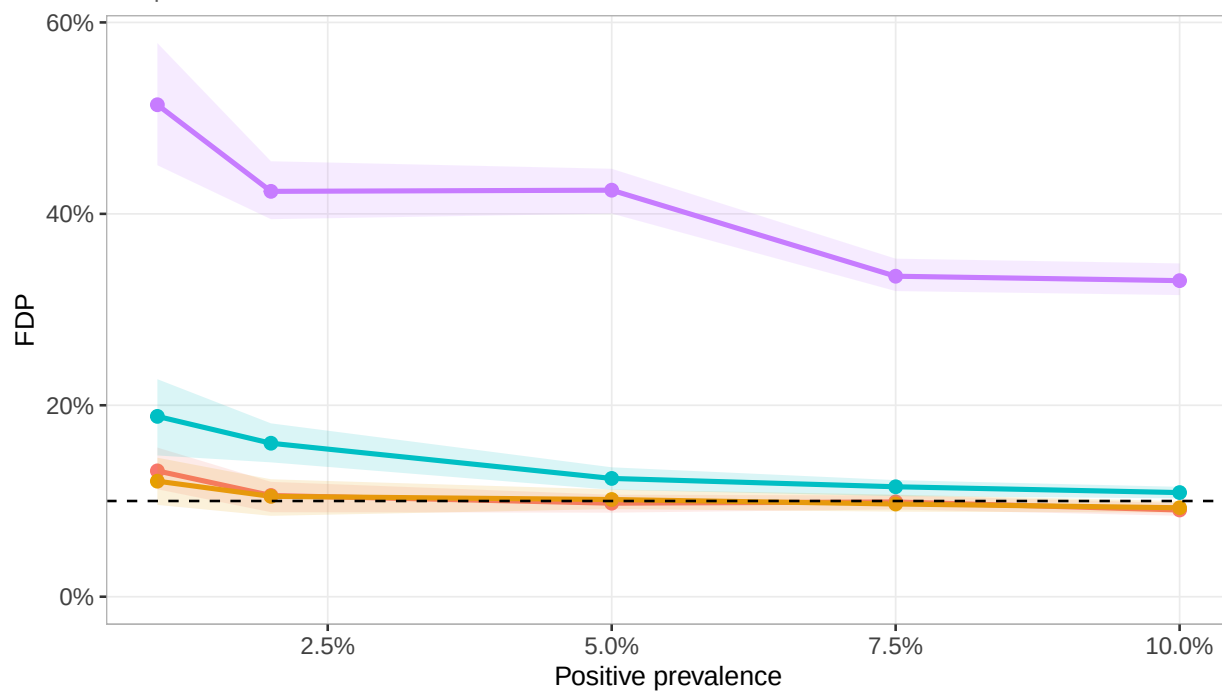
Replogle: #TP recovered vs #FP allowed



Method SCEPTRE SCEPTRE v0.3.0 Mixscale FR-Perturb

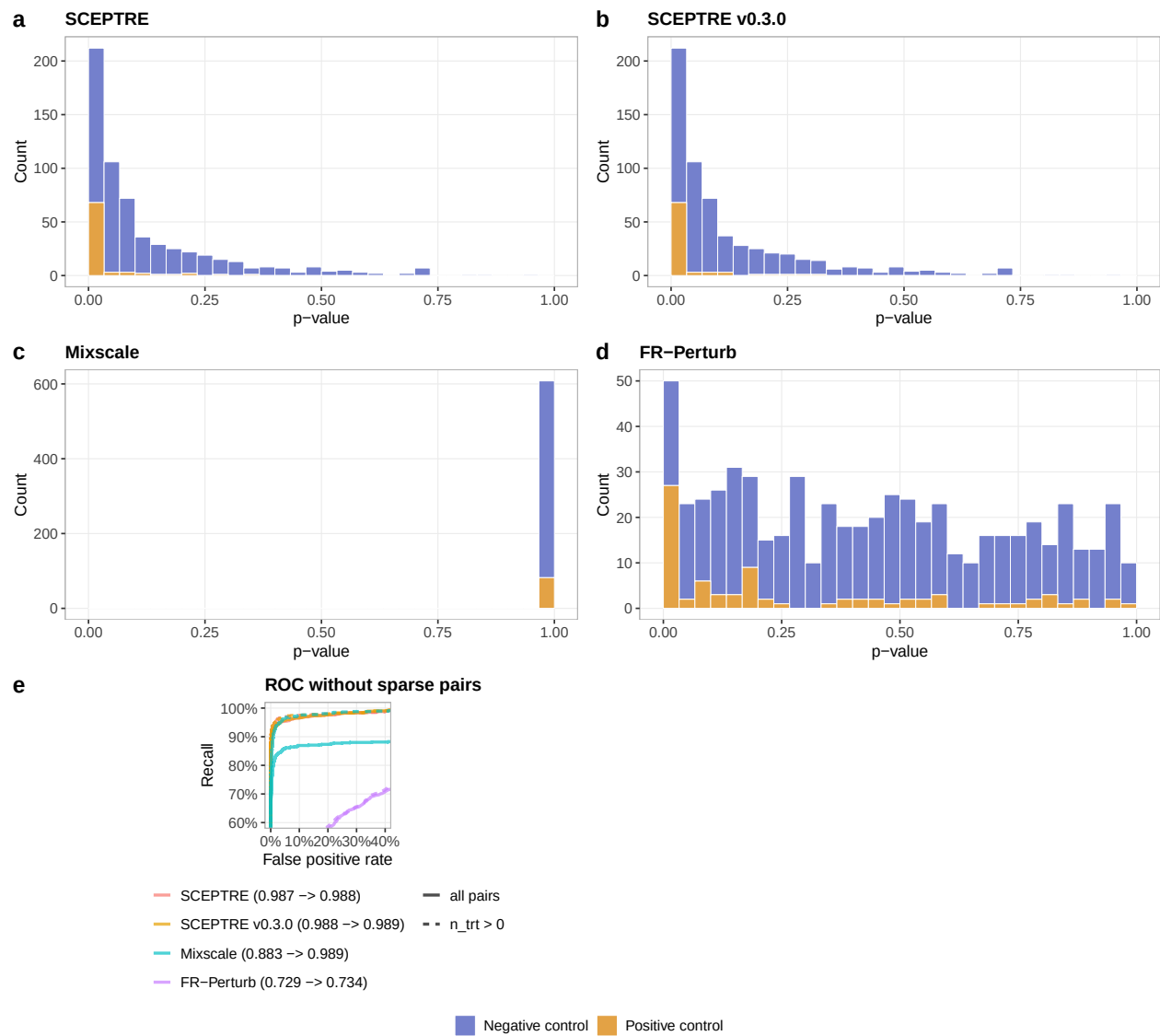
Replogle: FDP for varying positive prevalence

BH $q = 0.1$, $N = 7500$, $B = 100$



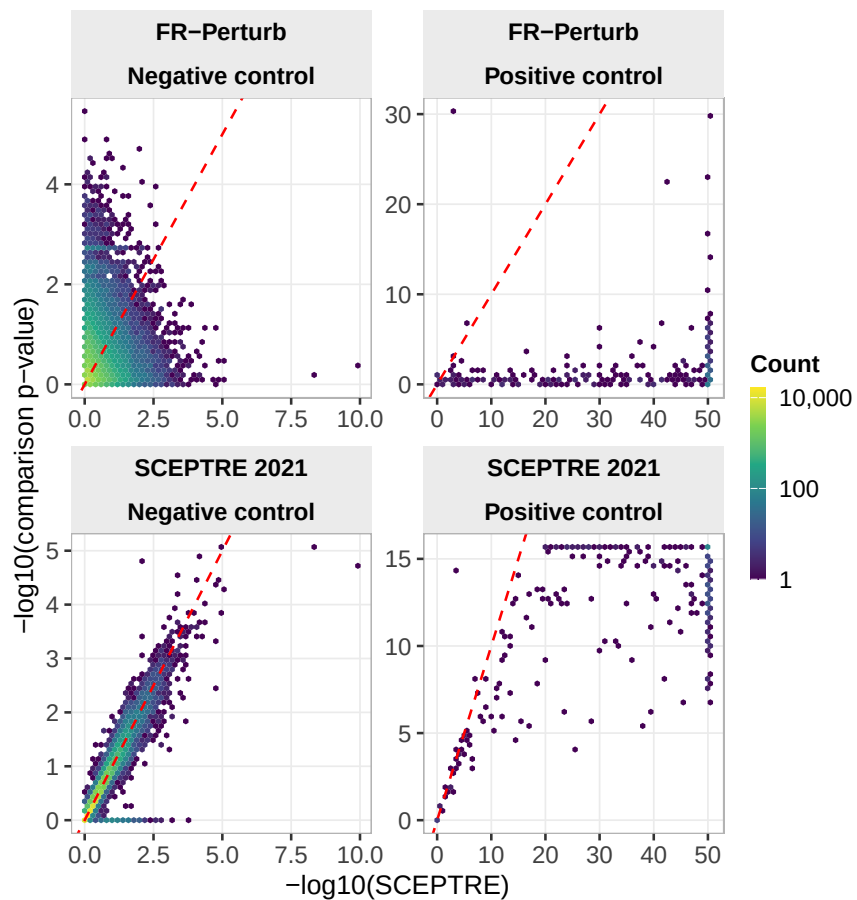
Method SCEPTRE SCEPTRE v0.3.0 Mixscale FR-Perturb

Reprogle: behavior for all-zero treatment groups

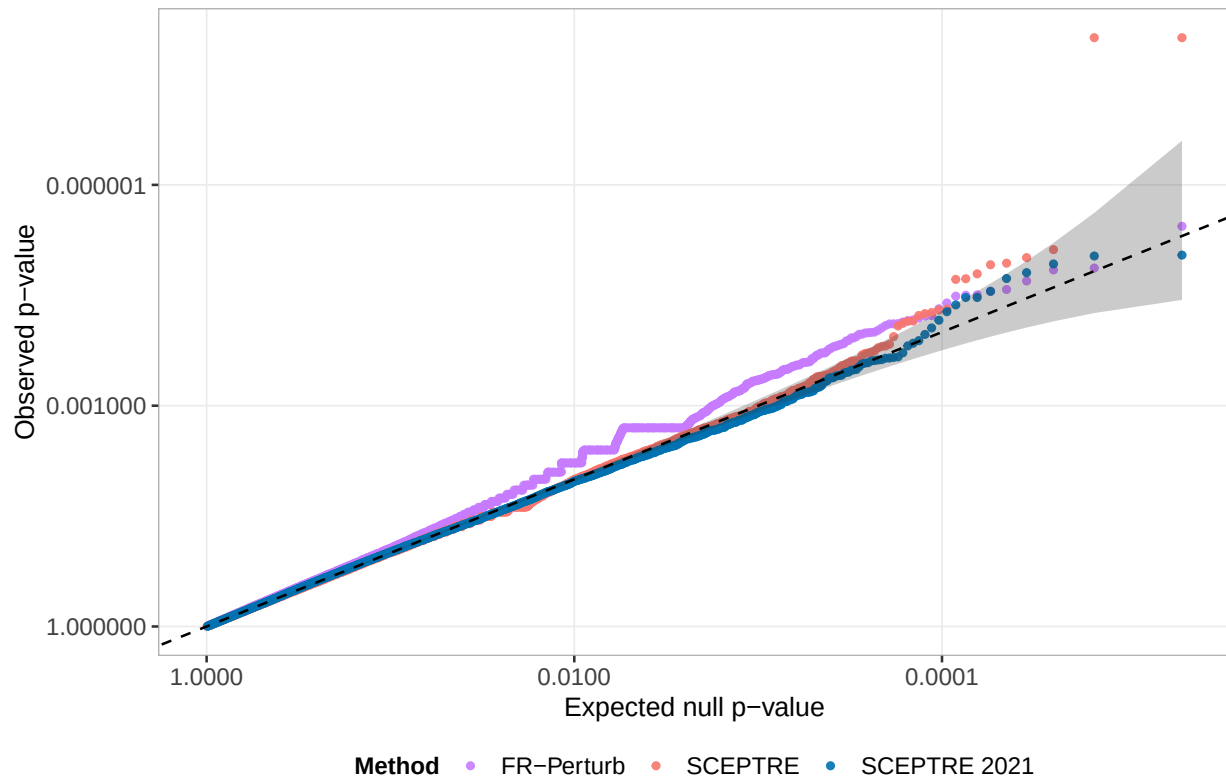


Gasperini: comparison with SCEPTRE manuscript from 2021

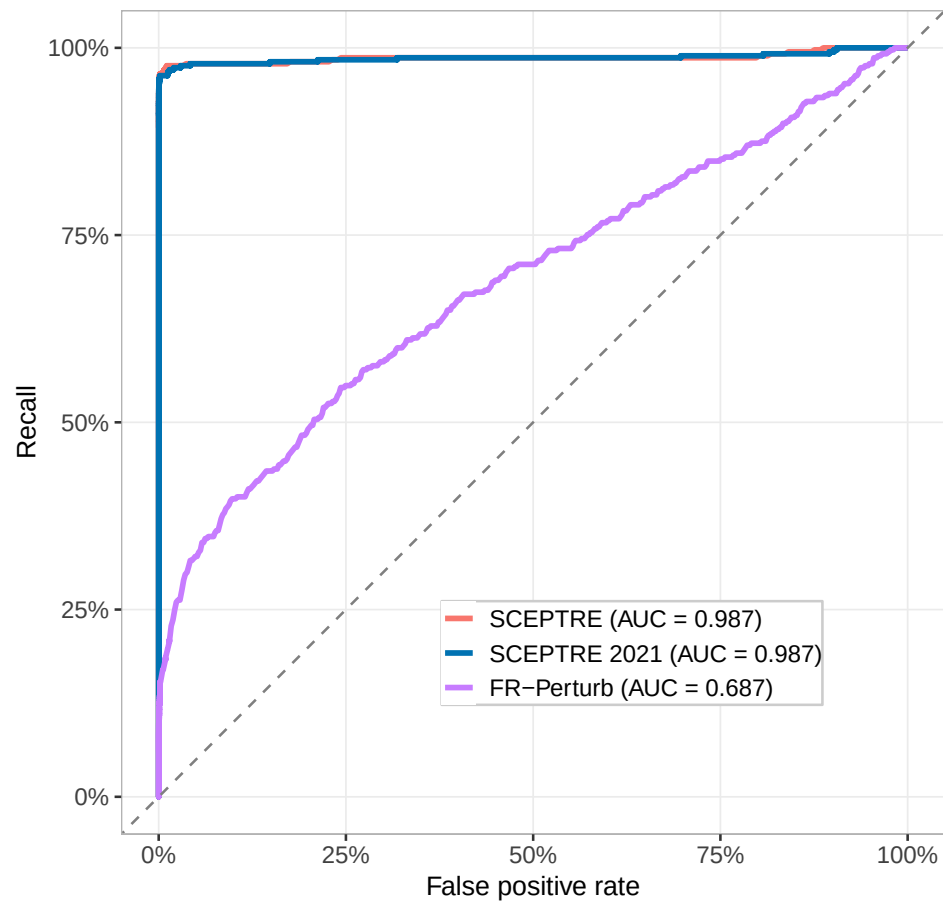
Gasperini: comparison of p-values by pair type

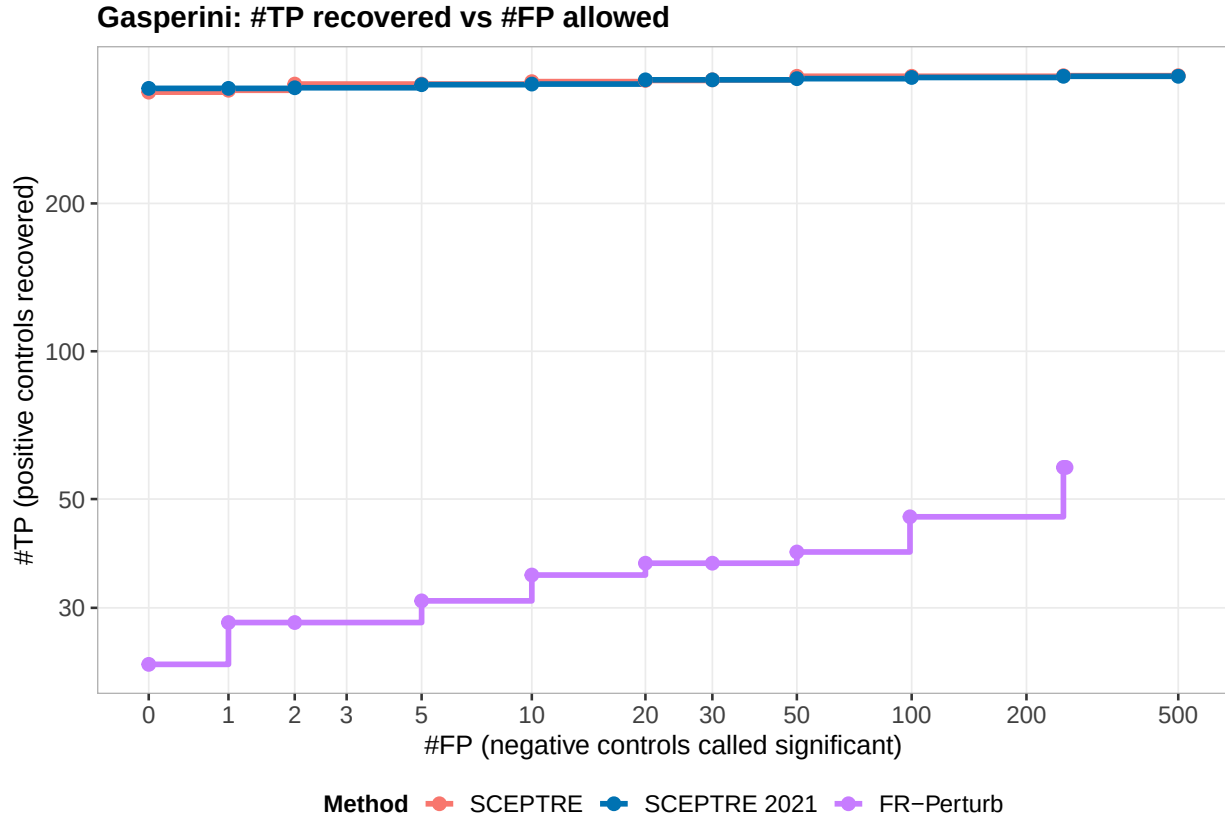


Gasperini: miscalibration in negative controls



Gasperini: separating positive and negative controls





computational runs

Assignment

The big thing I want to discuss is how I'm making computational benchmarking datasets.

We are going to use fishash's GuideBender model for computational benchmarking.

- Problem: varying N and G (number of cells and guides, respectively) separately leads to many implausible datasets. E.g. a low MOI dataset with $N = 10^3$ and $G = 10^4$ has around 10^{-1} perturbed cells per guide. No one is doing that experiment.
- Solution: we can avoid a lot of the bad (N, G) if we require a few properties to hold in our data. Let N_G^P be the typical number of actually perturbed cells for a guide. Note that we can count the total number of cells with guides in two ways:

$$G \times N_G^P = N \times \text{MOI}$$

Holding MOI fixed in our sequence is very reasonable. I think it's also reasonable to hold N_G^P fixed, i.e. we don't change the true positive sample size as we vary our dataset dimension. One of the reasons we didn't like downsampling cells is because that reduces the positive sample size. Then the above equation means that N/G is constant under these conditions, so rather than sweeping out a grid of (N, G) pairs, we sweep out a line with a fixed value of N/G . The hope is to get realistic small datasets by preserving these properties.

We do need to use an assignment algorithm to estimate N_G^P and MOI. I used thresholding with $T = 5$ for Gasperini and $T = 10$ for Replogle.

The final pieces we need are a couple GuideBender parameters:

1. SNR, the signal to noise ratio. This determines the fraction of total cell UMIs that are signal (i.e. coming from an actual perturbation) vs noise (either in-droplet or out-of-droplet error).

Association: computational benchmarking

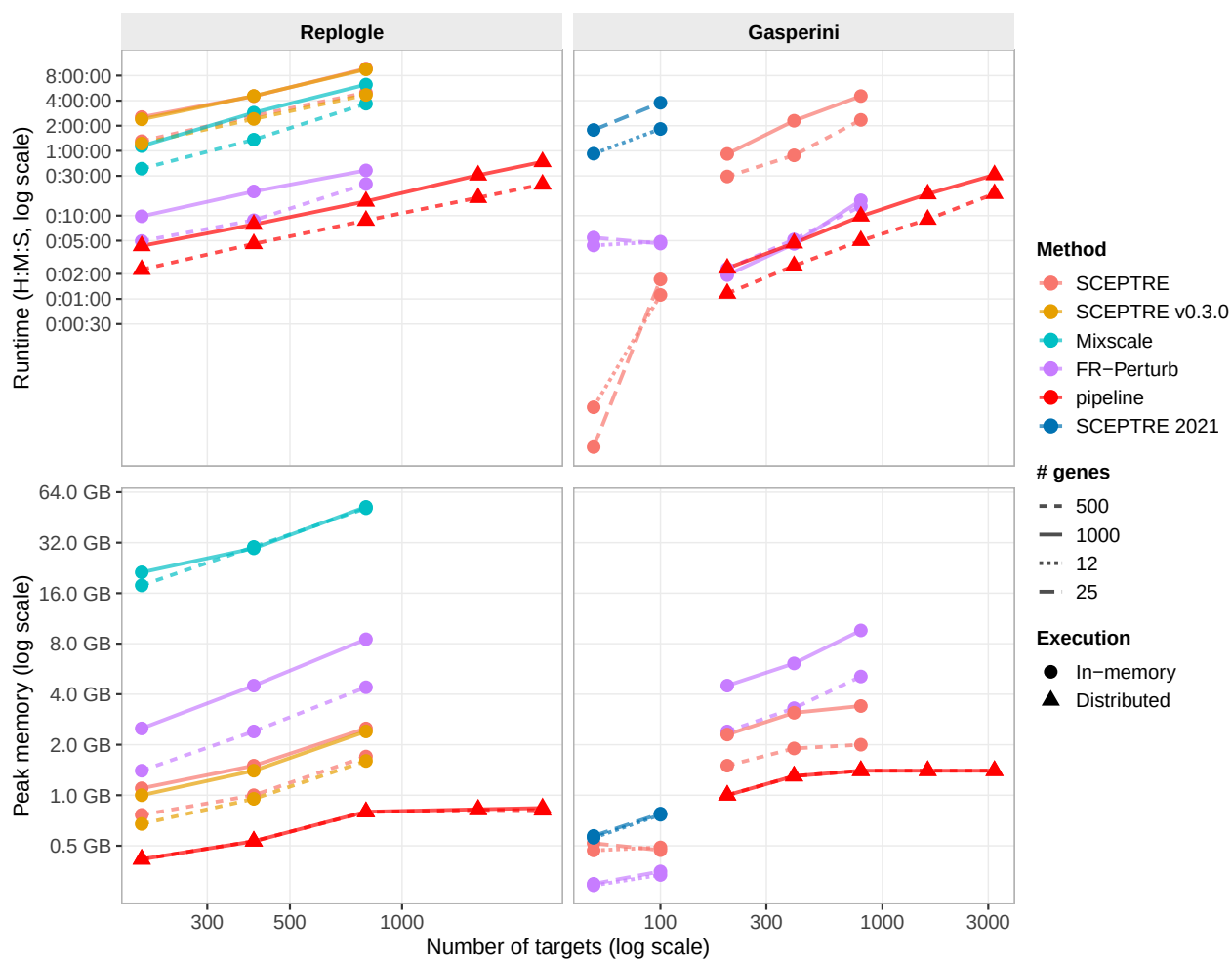


Figure 1: sceptre-manuscript is slow enough that I only ran it on some pretty small datasets, with 12 or 25 genes, and 50 or 100 targets. That's why Gasperini looks split. sceptre-v030 was able to run on the same datasets as the other methods.

2. **frac_endo** $\in [0, 1]$: this parameter determines what fraction of the noise UMIs are endogenous noise (in-droplet) vs exogenous noise (out-of-droplet)

SNR has consequences for the sparsity. Consider low MOI, where each guide contributes around N_G^P perturbed cells. Those will be mostly be non-zero, so the perturbed cells in this setup contribute a fixed number of non-zero entries to each guide. The rest of the non-zero entries come from the ambient noise. If the SNR is low, there will be many more UMIs allocated to the noise terms. This increases the chance of non-zero entries. On my Gasperini test run, I took $\text{SNR} \approx 20$ in order to get realistic sparsities. This could be a problem with this approach, but at least for computational purposes it's hopefully still usable. All other GuideBender parameters, including **frac_endo**, were kept at the defaults or values used by the fishash authors.

QUESTIONS

1. Does it make sense to keep the actual number of perturbed cells roughly constant per guide? In low MOI this makes sense, since each guide has a nearly disjoint number of cells expressing it, and it's reasonable to design an essentially balanced experiment where each guide contributes around the same number of cells. In high MOI too? Downsampling cells would change this property.
2. Does my use of thresholding to compute these seem suspect?
3. Is that Gasperini SNR pretty bad?